

Vertical AI

If you're reading this, then it would be fair for me to assume that you either use ChatGPT, Copilot, Claude or one of the many easily accessible and free-to-use services based on LLMs (Large Language Models). While these models are primarily trained on textual data, they have come to host a massive database of information across an equally massive collection of topics. You could ask these LLMs to write a poem or pen an essay or even solve mathematical problems, and they would easily finish the task. The output seems remarkably impressive until you start reading and then it dawns on you that the LLMs aren't always on the mark. The amount of hallucinations that these LLMs have is ridiculous. And unless you know about the domain, it becomes very difficult to spot the errors. There's a little bit of the Dunning Kruger effect in play here.

So why does this happen? Well, these models are trained on billions of parameters but that's far too less in reality. Let's say you have a model trained on general data. ChatGPT, for example, was trained on 570 GB of data at one point. This includes hundreds of thousands of books, web pages, transcripts, etc. Now imagine if you had an AI model that was trained on 570 GB or even much less than that, but this time, you have only fed it data from a particular domain, let's say finance. It knows how the biggest companies on the planet have cooked their books. All the different techniques that are used to save money. All the financial legislations in place in multiple countries. All the accounting software that was ever programmed. All the balance sheets ever filed by these companies. All the smarts that chartered accountants have applied to find loopholes. And so on. You now have an extremely proficient AI accountant.

This AI accountant can do everything. And it can work day-in and day-out without any rest. It is not subject to employee welfare regulations and does not need a weekly time off or compensation of days worked over time. It is faster than any accountant will ever be and it can handle the workload of an entire department, in parallel. At this point, you might as well do the AI a solid, and slap on the designation of Chief Financial Officer. It



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deserves it. What you have here, is an example of Vertical AI.

At its core, Vertical AI is the epitome of specialisation. Unlike its horizontal counterpart, which aims for broad applicability across various fields and domains, Vertical AI is akin to an ultra-specialised consultant. It is imbued with exhaustive knowledge and skills pertinent to a specific industry. And while my example was aimed at an entire department being run by one Vertical AI, it opens up to a wider application.

Imagine, a startup where the conventional team structure is replaced by a constellation of Vertical AI models, each championing a different facet of the business. You could have one for programming, one for UI/UX optimisation, one for market research and marketing, one for accounting and so on. Essentially, this paradigm offers a compelling narrative: a startup helmed by a sole founder who is augmented with several such Vertical AI models, each operating with the level of expertise that would have cost multiple crores to afford. This approach not only optimises operational efficiency but also democratises entrepreneurship.

This could very well be the future of the startup ecosystem with hundreds of thousands of startups sprouting with just one human being who is aided by a few Vertical AI models. Startups would be able to evolve or pivot at an unprecedented pace. New apps and services could be brought to life and made market ready within weeks, if not days.

The trajectory of Vertical AI underscores a transformative period in the intersection of technology and industry. As organisations embrace this paradigm, the delineation between human and machine work will blur. Thus, ushering in a new era of efficiency and innovation. However, this journey is accompanied by a tapestry of ethical, social and technical challenges. And we all know that an artificial general intelligence becomes the next logical step in such an evolution. While it is easy to assume that humanity, as we know it, will come to an end soon after, one shouldn't forget that this could very well be the thing that makes us a galaxy-faring civilisation. Did I stretch it too far there? You tell me. ■